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### Data management operations for data from edge locations

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#### Abstract

A method comprises receiving data from a plurality of devices and analyzing the data to identify one or more parameters for segregation of the data. In the method, the data is tagged with one or more metadata tags corresponding to the one or more parameters. The method further comprises segregating the data into a plurality of data portions based at least in part on the one or more parameters, and storing respective ones of the plurality of data portions in respective data stores of a plurality of data stores. The one or more parameters comprise at least one of one or more policies corresponding to the data and respective locations of the plurality of devices.

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## Background/Summary

## FIELD

(1) The field relates generally to information processing systems, and more particularly to data management in such information processing systems.

## BACKGROUND

(2) An edge computing architecture moves at least a portion of data processing to the periphery of a network to be closer to a data source rather than to a centralized location, e.g., data center. For example, instead of transmitting raw data to a data center to be processed and analyzed, such tasks or workloads are performed at or near locations where the data is actually generated. Edge computing architectures of this type can help to conserve network bandwidth, while also reducing latency and congestion, thus improving overall system performance.

(3) It is estimated that the amount of data being processed at edge locations will increase in upcoming years. With more data being processed at edge locations, the data will be stored in a distributed fashion, making it more difficult to access the data. Such difficulties are magnified in the case of device operational data, which may need to be accessed within strict timelines and with certain levels of security in order to address device issues in a timely and secure manner.

## SUMMARY

(4) Embodiments provide a data management platform in an information processing system.

(5) For example, in one embodiment, a method comprises receiving data from a plurality of devices and analyzing the data to identify one or more parameters for segregation of the data. In the method, the data is tagged with one or more metadata tags corresponding to the one or more parameters. The method further comprises segregating the data into a plurality of data portions based at least in part on the one or more parameters, and storing respective ones of the plurality of data portions in respective data stores of a plurality of data stores. The one or more parameters comprise at least one of one or more policies corresponding to the data and respective locations of the plurality of devices.

(6) Further illustrative embodiments are provided in the form of a non-transitory computer-readable storage medium having embodied therein executable program code that when executed by a processor causes the processor to perform the above steps. Still further illustrative embodiments comprise an apparatus with a processor and a memory configured to perform the above steps.

(7) These and other features and advantages of embodiments described herein will become more apparent from the accompanying drawings and the following detailed description.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 depicts an information processing system with a data management platform for controlling data access from edge locations in an illustrative embodiment.

(2) FIGS. 2A and 2B respectively depict edge and core/backend computing sites in an illustrative embodiment.

(3) FIG. 3 depicts a table illustrating data segregation based on different data requirements in an illustrative embodiment.

(4) FIG. 4 depicts a table illustrating regional data aggregation in an illustrative embodiment.

(5) FIG. 5 depicts a table illustrating data mapping to different communication networks in an illustrative embodiment.

(6) FIG. 6 depicts a process for controlling data access from edge locations according to an illustrative embodiment.

(7) FIGS. 7 and 8 show examples of processing platforms that may be utilized to implement at least a portion of an information processing system according to illustrative embodiments.

### DETAILED DESCRIPTION

(8) Illustrative embodiments will be described herein with reference to exemplary information processing systems and associated computers, servers, storage devices and other processing devices. It is to be appreciated, however, that embodiments are not restricted to use with the particular illustrative system and device configurations shown. Accordingly, the term “information processing system” as used herein is intended to be broadly construed, so as to encompass, for example, processing systems comprising cloud computing and storage systems, as well as other types of processing systems comprising various combinations of physical and virtual processing resources. An information processing system may therefore comprise, for example, at least one data center or other type of cloud-based system that includes one or more clouds hosting tenants that access cloud resources. Such systems are considered examples of what are more generally referred to herein as cloud-based computing environments. Some cloud infrastructures are within the exclusive control and management of a given enterprise, and therefore are considered “private clouds.” The term “enterprise” as used herein is intended to be broadly construed, and may comprise, for example, one or more businesses, one or more corporations or any other one or more entities, groups, or organizations. An “entity” as illustratively used herein may be a person or system. On the other hand, cloud infrastructures that are used by multiple enterprises, and not necessarily controlled or managed by any of the multiple enterprises but rather respectively controlled and managed by third-party cloud providers, are typically considered “public clouds.” Enterprises can choose to host their applications or services on private clouds, public clouds, and/or a combination of private and public clouds (hybrid clouds) with a vast array of computing resources attached to or otherwise a part of the infrastructure. Numerous other types of enterprise computing and storage systems are also encompassed by the term “information processing system” as that term is broadly used herein.

(9) As used herein, “real-time” refers to output within strict time constraints. Real-time output can be understood to be instantaneous or on the order of milliseconds or microseconds. Real-time output can occur when the connections with a network are continuous and a user device receives messages without any significant time delay. Of course, it should be understood that depending on the particular temporal nature of the system in which an embodiment is implemented, other appropriate timescales that provide at least contemporaneous performance and output can be achieved.

(10) FIG. 1 shows an information processing system **100** configured in accordance with an illustrative embodiment. The information processing system **100** comprises user devices **102-1**, . . . **102-M** (collectively “user devices **102**”) including respective instances of device management logic **103-1**, . . . **103-M** (collectively “device management logic **103**”). The information processing system **100** further comprises at least one core/backend computing site **106** and dashboards and application programming interfaces (APIs) **109** corresponding to the at least one core/backend computing site **106**. The user devices **102** communicate over the network **104** (which may comprise multiple networks) with a data management platform **110**, and with a core/backend computing site **106**. The variable M and other similar index variables herein such as K, L and P are assumed to be arbitrary positive integers greater than or equal to one.

(11) The user devices **102** can comprise, for example, desktop, laptop or tablet computers, servers, host devices, storage devices, switches, mobile telephones, Internet of Things (IoT) devices or other types of processing devices capable of communicating with the data management platform **110** over the network **104**. Such devices are examples of what are more generally referred to herein as “processing devices.” Some of these processing devices are also generally referred to herein as “computers.” The user devices **102** may also or alternately comprise virtualized computing resources, such as virtual machines (VMs), containers, etc. The user devices **102** in some embodiments comprise respective computers associated with a particular company, organization or other enterprise.

(12) The terms “user” or “administrator” herein are intended to be broadly construed so as to

encompass numerous arrangements of human, hardware, software or firmware entities, as well as combinations of such entities. At least a portion of the available services and functionalities provided by the data management platform **110** in some embodiments may be provided under Function-as-a-Service (“FaaS”), Containers-as-a-Service (“CaaS”) and/or Platform-as-a-Service (“PaaS”) models, including cloud-based FaaS, CaaS and PaaS environments.

(13) Although not explicitly shown in FIG. **1**, one or more input-output devices such as keyboards, displays or other types of input-output devices may be used to support one or more user interfaces to the data management platform **110**, as well as to support communication between the data management platform **110**, connected devices (e.g., user devices **102**), connected sites (e.g., core/backend computing sites **106**) and/or other related systems and devices not explicitly shown.

(14) In some embodiments, the user devices **102** are assumed to be associated with repair and/or support technicians, system administrators, information technology (IT) managers, software developers, release management personnel or other authorized personnel configured to access and utilize the data management platform **110**.

(15) In some embodiments, the user devices **102** may be part of an edge computing site as “edge devices.” An edge computing site comprises one or more edge stations or other types and arrangements of edge nodes illustratively comprising a plurality of edge devices. Referring for example to FIG. **2A**, an example edge computing site **205** comprises a set of edge compute, storage and/or network resources **207**. A given such set of edge resources illustratively comprises at least one of compute, storage and network resources of one or more edge devices of the corresponding edge computing site.

(16) Referring to FIG. **2B**, an example core/backend computing site **206** (which may be the same or similar to core/backend computing site **106**) comprises a set of core compute, storage and/or network resources **208**. The core/backend computing site **106/206** comprises one or more data centers or other types and arrangements of core/backend nodes. The core/backend computing site **106/206** illustratively comprises at least one data center implemented at least in part utilizing cloud infrastructure. Each such node of an edge computing site **205**, backend/core computing site **106/206** or other computing site comprises at least one processing device that includes a processor coupled to a memory.

(17) The edge computing site **205** and the core/backend computing site **106/206** illustratively execute at least portions of various workloads for system users. Such workloads may comprise one or more applications. As used herein, the term “application” is intended to be broadly construed to encompass, for example, microservices and other types of services implemented in software executed by the sites **205** or **106/206**. Such applications can include edge-hosted applications running on the edge computing site **205** and core/backend-hosted applications running on the core/backend computing site **106/206**.

(18) Edge and core compute resources of the edge and core/backend computing sites **205** and **106/206** can include, for example, various arrangements of processors, possibly including associated accelerators. Edge and core storage resources of the edge and core/backend computing sites **205** and **106/206** can include, for example, one or more storage systems or portions thereof that are part of or otherwise associated with the edge and core/backend computing sites **205** and **106/206**. A given such storage system may comprise, for example, all-flash and hybrid flash storage arrays, software-defined storage systems, cloud storage systems, object-based storage systems, and scale-out distributed storage clusters. Combinations of multiple ones of these and other storage types can also be used in implementing a given storage system in an illustrative embodiment. Edge and core network resources of the edge and core/backend computing sites **205** and **106/206** can include, for example, resources of various types of network interface devices providing particular bandwidth, data rate and communication protocol features.

(19) An edge computing site **205** comprises a plurality of edge devices, with a given such edge device (e.g., user device **102**) comprising a processing device that includes a processor coupled to a

memory. One or more core/backend computing sites **106/206** may comprise, for example, at least one data center implemented at least in part utilizing cloud infrastructure. It is to be appreciated, however, that illustrative embodiments disclosed herein do not require the use of cloud infrastructure. The one or more edge and core/backend computing sites **205** and **106/206** are each assumed to be implemented using at least one processing device of at least one processing platform. Each such processing device generally comprises at least one processor and an associated memory, and implements at least a portion of the functionality of the edge and core compute, storage and/or network resources. Such processing devices can illustratively include particular arrangements of compute, storage and network resources.

(20) As noted herein above, with more data being processed at edge locations, the data will be stored in a distributed fashion, making it more difficult to access the data. Illustrative embodiments provide techniques to address the difficulties in accessing data from edge locations. For example, the embodiments provide a framework which segregates data (e.g., operational data) based on one or more parameters so that critical and important data can be accessed and processed more quickly than other types of data. Advantageously, the embodiments provide technical solutions to segregate the data in secure logical data stores and to aggregate and correlate the data in response incoming queries.

(21) The data management platform **110** in the present embodiment is assumed to be accessible to the user devices **102**, core/backend computing site **106** and vice versa over the network **104**. The network **104** is assumed to comprise a portion of a global computer network such as the Internet, although other types of networks can be part of the network **104**, including a wide area network (WAN), a local area network (LAN), a satellite network, a telephone or cable network, a cellular network such as, for example, fourth generation (4G) (e.g., 4G LTE) or fifth generation (5G) (e.g., 5G stand-alone (5G SA), 5G non-stand-alone (5G NSA)), a private network (e.g., Private Long-Term Evolution (pLTE)), a wireless network such as a WiFi or WiMAX network, or various portions or combinations of these and other types of networks. As explained in more detail herein, the network **104** in some embodiments therefore comprises combinations of multiple different types of networks each comprising processing devices configured to communicate using Internet Protocol (IP) or other related communication protocols.

(22) As a more particular example, some embodiments may utilize one or more high-speed local networks in which associated processing devices communicate with one another utilizing Peripheral Component Interconnect express (PCIe) cards of those devices, and networking protocols such as InfiniBand, Gigabit Ethernet or Fibre Channel. Numerous alternative networking arrangements are possible in a given embodiment, as will be appreciated by those skilled in the art.

(23) Referring to FIG. **1**, the data management platform **110** includes a network connectivity engine **120**, a support services engine **130** and a data management engine **140**. The data management platform **110** runs on an underlying operating system (OS) **150** such as, for example, a Windows® or Linux® OS. The network connectivity engine **120** includes a policy application layer **121**, an orchestration layer **122**, an adapter layer **123**, a data collection layer **124**, a data store **125** and a cache **126**. The support services engine **130** includes a connection layer **131**, an inventory layer **132**, a cases layer **133**, an operational data layer **134** and a data store **135**. The data management engine **140** includes a policy enforcement layer **141**, an administration layer **142**, a classification layer **143**, a segregation and aggregation layer **144**, a data store **145** and a mapping layer **146**. In some embodiments, the data management platform **110** or a portion thereof is an element of an edge computing site **205**. Additionally, although one data management platform **110** is shown, there may be multiple data management platforms **110** that may be elements of respective edge computing sites **205**.

(24) The instances of device management logic **103** in the user devices **102** can be configured to collect data from the user devices **102**. In illustrative embodiments, the data collected includes operational data comprising information identifying the user devices **102** and the components of

the user devices **102**, information corresponding to alerts about the health and operational status of the components of the user devices **102**, information corresponding to applications running on the user devices **102** and information corresponding to workloads from the user devices **102** that need to be processed by edge and/or core/backend computing sites **205** and **106/206**. As used herein, a “component” is to be broadly construed, and can refer to various parts, hardware components and/or software components such as, but not necessarily limited to, storage devices (e.g., hard disk drives), batteries, chassis, display panels, motherboards, central processing units (CPUs), controllers, cards, heat sinks, fans, fan assemblies, processors, ports, port connectors, host bus adaptors (HBAs), speakers, keyboards, memories, servers, switches, sensors, buses (e.g., serial buses) or other elements of a user device **102**.

(25) In an illustrative embodiment, the data collected by the instances of device management logic **103** is received and/or retrieved by the data management platform **110** via the connection layer **131** of the support services engine **130** and provided to the data collection layer **124** of the network connectivity engine **120** and to the administration layer **142** of the data management engine **140**. In illustrative embodiments, the data received and/or retrieved via the connection layer **131** is sent to the inventory layer **132**, which classifies the data based on origin (e.g., user device **102**, component, application, etc.), and sends the classified data to the data collection layer **124**. The data received by the data collection layer **124** is stored in at least one data store **125**.

(26) The data received and/or retrieved by the data management platform **110** is received over, for example, a first network, such as a wired and/or wireless (e.g., WiFi or WiMAX) network. In illustrative embodiments, the policy enforcement layer **141** of the data management engine **140** analyzes the data (e.g., operational data) to identify one or more parameters for segregation of the data. The one or more parameters comprise, for example, one or more policies corresponding to the data and/or respective locations of the plurality of devices. For example, the one or more policies specify one or more data transmission requirements such as, but not necessarily limited to, latency, bandwidth, security and/or data gravity. For example, some of the applications running on the user devices **102** may specify certain data transmission requirements such as, but not necessarily limited to, high bandwidth and low latency, a secure network and low bandwidth, etc. In some cases, an application may not specify or require one or more of the parameters. Such requirements and/or specifications may be identified, for example, in metadata associated with an application, in commands or instructions associated with an application, in application code, and/or in service level agreements (SLAs) accessible to the policy enforcement layer **141**. The SLAs may be associated with a user device **102** and/or application.

(27) In illustrative embodiments, a charging function, which may be part of the policy enforcement layer **141**, defines critical values by assigning weights to input variables such as, for example, required latency, bandwidth, security, data gravity, etc. For example, some parameters may be assigned higher weights based on whether those parameters are more crucial to the processing of an application workload than other parameters. For example, in the case of video processing applications, high bandwidth may be weighted higher than other factors, in the case of applications transmitting sensitive information, network security may be weighted higher than other factors and in the case intrusion detection applications, low latency may be weighted higher than other applications.

(28) The locations of the plurality of devices correspond to IP geolocations. In more detail, the IP address of a given device (e.g., one of the user devices **102**) is recorded by the data collection layer **124**, and the geographic location of the given device is identified based on the IP address. The policy enforcement layer **141** comprises, for example, a geolocation API, which is used to identify the geographic locations.

(29) The classification layer **143** of the data management engine **140** tags the data with one or more metadata tags corresponding to the one or more parameters that have been identified by the policy enforcement layer **141**. In addition, the classification layer **143** classifies the data into one or more

functional groups. Each functional group corresponds to, for example, commonly defined actions that can be performed by the functional group. Some non-limiting examples of functional groups are “video streaming,” “secure banking,” “intrusion detection,” etc.

(30) The segregation and aggregation layer **144** segregates the data into a plurality of data portions based at least in part on the one or more parameters and/or the one or more functional groups. The segregated data portions are stored in respective data stores based on how the data portions have been segregated.

(31) In a non-limiting operational example, referring to the table **300** in FIG. **3**, operational data from four devices A, B, C and D (e.g., user devices **102**) is received by the data management platform **110**, and the policy enforcement layer **141** identifies two policies: “high bandwidth” and “low latency” from the data. In this case, devices A and C require high bandwidth, and devices B and D require low latency. The policy enforcement layer **141** determines that device A and C are in region I, and devices B and D are in region II. The segregation and aggregation layer **144** segregates the operational data based on policy at each region. In this example, a data store in region I stores data from devices A and C, and a data store in region II stores data from devices B and D. The data store **145**, which may be part of an edge computing site (e.g., edge computing site **205**) may be one of the data stores used for storing segregated data. Alternatively, one or more data stores in edge computing sites corresponding to region I and/or region II can be used to store the segregated data.

(32) In addition, the classification layer **143** classifies the data corresponding to devices A and C in the video streaming functional group, and the data corresponding to devices B and D in the secure banking functional group. The classification layer **143** tags with, for example, geo location metadata (region I or II), policy type metadata and functional group metadata. For example, data from devices A and C includes “high bandwidth,” “region I” and “video streaming” tags, and data from devices B and D includes “low latency,” “region II” and “secure banking” tags. Certain data stores may be limited to data corresponding to a designated policy (e.g., high bandwidth or low latency), a given location (e.g., region I or II) and/or a particular functional group (e.g., video streaming or secure banking).

(33) The policy enforcement layer **141** further assigns access to users to respective ones of the data portions based at least in part on roles of the users. For example, referring to the table **300**, users with a video administration role are provided with access to data portions corresponding to devices A and C and high bandwidth in the video streaming functional group, and not to data portions corresponding to devices B and D. Users with a banking analysis role are provided with access to data portions corresponding to devices B and D and low latency in the secure banking functional group, and not to data portions corresponding to device A and C.

(34) Although data may be segregated, the illustrative embodiments provide techniques to allow for aggregated data monitoring and operations across distributed data stores. For example, the segregation and aggregation layer **144** aggregates segregated data from distributed data stores in different edge locations based on, for example, common parameters and/or user access rights. In a non-limiting illustrative embodiment, the segregation and aggregation layer **144** performs logical aggregation of data from multiple edge locations based on the data falling under the same policies and/or access rights of users. The data is tagged based on, for example, policies allowing access and limiting user operations.

(35) For example, referring to the table **400** in FIG. **4**, video streaming data can be aggregated across two regions for billing. Through aggregation, multiple data stores in different edge locations (region I and II) whose data fall under the same functional group (video streaming) are accessible to tenants with access to this functional group and the particular regions. In this case users who are billing managers with a data store financials role are given access to data from both regions I and II so that reports for clients A and B can be generated. The reports include a client A bill for region I, a client B bill for region I, a client A bill for region II, and a client B bill for region II.



(36) As can be understood, the segregation and aggregation layer **144** aggregates data portions from respective data stores based at least in part on one or more common parameters, and the policy enforcement layer **141** enables access by at least one user to the aggregated data portions. In illustrative embodiments, the aggregating is an automated response to one or more queries for the data from users with access rights to the data. In some embodiments, a core/backend computing site **106** via the dashboards and APIs **109**, generates a visualization of the aggregated data portions from respective data stores in different edge locations. The aggregated data portions may be received by the core/backend computing site **106** from the respective data stores in a plurality of data streams. The APIs correlate multiple data streams of, for example, operational data, to deliver real-time insights in an integrated dashboard. In illustrative embodiments, the aggregating of the data portions is based at least in part on the assigned access of users to the data portions, which as noted herein, may be based at least in part on a role of each user.

(37) In illustrative embodiments, the mapping layer **146** of the of the data management engine **140** maps segregated data portions to one or more network slices through which the data of the data portions can be transmitted from data stores at different edge locations to the core/backend computing site **106**. An example data store from which transmission may occur is data store **145**, and an example edge location is edge computing site **205**. The network slices may be of 4G network, a 5G network and/or a pLTE network. The network slices are determined based at least in part on one or more parameters (e.g., high bandwidth, low latency, etc.).

(38) In illustrative embodiments, the policy application layer **121** in conjunction with the mapping layer **146** analyzes the data portions to identify one or more conditions for transmission of the data portions to the core/backend computing site **106**. The one or more conditions for transmission of the data comprise, but are not necessarily limited to latency, bandwidth, security and/or data gravity. Based, at least in part, on the one or more conditions, the policy application layer **121** and the mapping layer determine a network to be utilized for transmission of the data to the core/backend computing site **106**. The orchestration layer **122** and administration layer **142** in conjunction with the policy application layer **121** and mapping layer **146** assign the data portions to a particular network and/or a particular portion of a network (e.g., network slice). As used herein, “network slices” or “network slicing” is to be broadly construed to refer to, for example, a network architecture enabling multiplexing of virtualized and independent logical networks on the same physical network infrastructure. A network slice may be an isolated end-to-end network tailored to fulfill the requirements of a given application. Some non-limiting examples of network slices include, for example, an ultra-reliable low latency connection (URLLC) network slice, an enhanced mobile broadband (eMBB) network slice and a massive machine-to-machine type communication (mMTC) network slice, which may correspond to IoT traffic. For example, augmented reality video traffic may require both eMBB (high bandwidth) and URLLC (low latency) characteristics.

(39) The mapping layer **146** logically maps data portions onto domains based on, for example, policies. The data portions may be tagged and classified based on policies to form logical separate domains extending from edge to backend locations. When connected over a designated communication network, data policies are enforced to map the connection over the right network slice to meet data requirements. The appropriate mapping also allows for SLA requirements of the data and user to be met.

(40) For example, referring to the table **500** in FIG. **5**, video streaming and secure banking data can be mapped to meet end user requirements of high bandwidth and low latency. For example, the high bandwidth policy is mapped to an eMBB network slice, and the low latency policy is mapped to a URLLC network slice. Device A, which is from the video streaming functional group, corresponds to the high bandwidth policy. Data from device A is transmitted via the eMBB network slice to the core/backend computing site **106** via a 5G core from region I. Device B, which is from the secure banking functional group, corresponds to the low latency policy. Data from device B is transmitted via the URLLC network slice to the core/backend computing site **106** via a 5G core

from region II. The region I data store is located closer to device A than device B and is distinct from the region II data store, which is located closer to device B than device A. Data portions that are queued for transmission may be stored in cache **126** before being transmitted over a selected network or network slice.

(41) Referring back to FIG. **1**, user devices **102** send data to the support services engine **130** via the connection layer **131**. The data is sent to an inventory layer **132** to classify the data, for example, based on origin. A cases layer **133** processes and aggregates the data to be utilized for support services and/or technical support cases. The operational data layer **134** filters the aggregated data for operational details, and the filtered data is stored in a data store **135** for future consumption by, for example a core/backend computing site **106** of an enterprise managing the user devices **102**.

(42) The adapter layer **123** of the network connectivity engine **120** and the administration layer **142** of the data management engine **140** are utilized for communication and protocol integration. For example, the adapter layer **123** communicates with the connection layer **131** of the support services engine **130** to allow the support services engine **130** to interoperate with the user devices **102** regardless of their corresponding connectivity protocols. For example, a driver in the adapter layer **123** translates a variety of connectivity protocols associated with, for example, 4G and/or 5G networks, to a connectivity protocol used for broadband connections such as, but not necessarily limited to Internet Protocol over Ethernet (IPoE) or point-to-point over Ethernet (PPPoE) protocols.

(43) In addition, once a network or network slice has been selected based on one or more of the requirements discussed herein above, the adapter layer **123** and the administration layer **142** configure data for transmission over a selected network or network slice. For example, the adapter layer **123** and the administration layer **142** identify one or more connectivity protocol types in one or more packet fields of the data, and execute one or more interfaces for parsing the data to conform with the one or more connectivity protocols used by the selected network or network slice so that the data can be sent to edge and/or core/backend computing sites **205** and **106/206** over the selected network or network slice. Data may be stored in cache **126** before being transmitted over the selected network or network slice.

(44) The embodiments discussed herein may be used in connection with user devices **102** operating over a 5G network and requiring quick interpretation and action on the data. Such user devices **102** may be used in connection with, for example, robotics in manufacturing scenarios to enhance cognitive quality and distributed processing of data closer to or at edge locations. In another non-limiting example, the illustrative embodiments may be used in connection with securing access to sensitive information such as, for example, personal General Data Protection Regulation (GDPR) data and salary information, which may require authorized users and users in certain geographical areas to access the information.

(45) According to one or more embodiments, the data stores **125**, **135** and **145**, cache **126** and other data stores, repositories or databases referred to herein can be configured according to a relational database management system (RDBMS) (e.g., PostgreSQL). In some embodiments, the data stores **125**, **135** and **145**, cache **126** and other data stores, repositories or databases referred to herein are implemented using one or more storage systems or devices associated with the data management platform **110**. In some embodiments, one or more of the storage systems utilized to implement the data stores **125**, **135** and **145**, cache **126** and other data stores, repositories or databases referred to herein comprise a scale-out all-flash content addressable storage array or other type of storage array.

(46) The term “storage system” as used herein is therefore intended to be broadly construed, and should not be viewed as being limited to content addressable storage systems or flash-based storage systems. A given storage system as the term is broadly used herein can comprise, for example, network-attached storage (NAS), storage area networks (SANs), direct-attached storage (DAS) and distributed DAS, as well as combinations of these and other storage types, including software-defined storage.

(47) Other particular types of storage products that can be used in implementing storage systems in illustrative embodiments include all-flash and hybrid flash storage arrays, software-defined storage products, cloud storage products, object-based storage products, and scale-out NAS clusters. Combinations of multiple ones of these and other storage products can also be used in implementing a given storage system in an illustrative embodiment.

(48) Although shown as elements of the data management platform **110**, the network connectivity engine **120**, support services engine **130** and/or data management engine **140** in other embodiments can be implemented at least in part externally to the data management platform **110**, for example, as stand-alone servers, sets of servers or other types of systems coupled to the network **104**. For example, the network connectivity engine **120**, support services engine **130** and/or data management engine **140** may be provided as cloud services accessible by the data management platform **110**.

(49) The network connectivity engine **120**, support services engine **130** and/or data management engine **140** in the FIG. 1 embodiment are each assumed to be implemented using at least one processing device. Each such processing device generally comprises at least one processor and an associated memory, and implements one or more functional modules for controlling certain features of the network connectivity engine **120**, support services engine **130** and/or data management engine **140**.

(50) At least portions of the data management platform **110** and the elements thereof may be implemented at least in part in the form of software that is stored in memory and executed by a processor. The data management platform **110** and the elements thereof comprise further hardware and software required for running the data management platform **110**, including, but not necessarily limited to, on-premises or cloud-based centralized hardware, graphics processing unit (GPU) hardware, virtualization infrastructure software and hardware, Docker containers, networking software and hardware, and cloud infrastructure software and hardware.

(51) Although the network connectivity engine **120**, support services engine **130**, data management engine **140** and other elements of the data management platform **110** in the present embodiment are shown as part of the data management platform **110**, at least a portion of the network connectivity engine **120**, support services engine **130**, data management engine **140** and other elements of the data management platform **110** in other embodiments may be implemented on one or more other processing platforms that are accessible to the data management platform **110** over one or more networks. Such elements can each be implemented at least in part within another system element or at least in part utilizing one or more stand-alone elements coupled to the network **104**.

(52) It is assumed that the data management platform **110** in the FIG. 1 embodiment and other processing platforms referred to herein are each implemented using a plurality of processing devices each having a processor coupled to a memory. Such processing devices can illustratively include particular arrangements of compute, storage and network resources. For example, processing devices in some embodiments are implemented at least in part utilizing virtual resources such as virtual machines (VMs) or Linux containers (LXCs), or combinations of both as in an arrangement in which Docker containers or other types of LXCs are configured to run on VMs.

(53) The term “processing platform” as used herein is intended to be broadly construed so as to encompass, by way of illustration and without limitation, multiple sets of processing devices and one or more associated storage systems that are configured to communicate over one or more networks.

(54) As a more particular example, the network connectivity engine **120**, support services engine **130**, data management engine **140** and other elements of the data management platform **110**, and the elements thereof can each be implemented in the form of one or more LXCs running on one or more VMs. Other arrangements of one or more processing devices of a processing platform can be used to implement the network connectivity engine **120**, support services engine **130** and data management engine **140**, as well as other elements of the data management platform **110**. Other

portions of the system **100** can similarly be implemented using one or more processing devices of at least one processing platform.

(55) Distributed implementations of the system **100** are possible, in which certain elements of the system reside in one data center in a first geographic location while other elements of the system reside in one or more other data centers in one or more other geographic locations that are potentially remote from the first geographic location. Thus, it is possible in some implementations of the system **100** for different portions of the data management platform **110** to reside in different data centers. Numerous other distributed implementations of the data management platform **110** are possible.

(56) Accordingly, one or each of the network connectivity engine **120**, support services engine **130**, data management engine **140** and other elements of the data management platform **110** can each be implemented in a distributed manner so as to comprise a plurality of distributed elements implemented on respective ones of a plurality of compute nodes of the data management platform **110**.

(57) It is to be appreciated that these and other features of illustrative embodiments are presented by way of example only, and should not be construed as limiting in any way. Accordingly, different numbers, types and arrangements of system elements such as the network connectivity engine **120**, support services engine **130**, data management engine **140** and other elements of the data management platform **110**, and the portions thereof can be used in other embodiments.

(58) It should be understood that the particular sets of modules and other elements implemented in the system **100** as illustrated in FIG. **1** are presented by way of example only. In other embodiments, only subsets of these elements, or additional or alternative sets of elements, may be used, and such elements may exhibit alternative functionality and configurations.

(59) For example, as indicated previously, in some illustrative embodiments, functionality for the data management platform can be offered to cloud infrastructure customers or other users as part of FaaS, CaaS and/or PaaS offerings.

(60) The operation of the information processing system **100** will now be described in further detail with reference to the flow diagram of FIG. **6**. With reference to FIG. **6**, a process **600** for controlling data access from edge locations as shown includes steps **602** through **610**, and is suitable for use in the system **100** but is more generally applicable to other types of information processing systems comprising a data management platform configured for controlling data access from edge locations.

(61) In step **602**, data is received from a plurality of devices. In illustrative embodiments, the data comprises operational data of the plurality of devices. In step **604**, the data is analyzed to identify one or more parameters for segregation of the data. The one or more parameters comprise, for example, one or more policies corresponding to the data and/or respective locations of the plurality of devices.

(62) In step **606**, the data is tagged with one or more metadata tags corresponding to the one or more parameters. In step **608**, the data is segregated into a plurality of data portions based at least in part on the one or more parameters. In step **610**, respective ones of the plurality of data portions are stored in respective data stores of a plurality of data stores. At least a subset of the respective data stores are in respective edge locations corresponding to the plurality of devices.

(63) In illustrative embodiments, the plurality of data portions are mapped to one or more network slices of a 5G network and/or a pLTE network based at least in part on the one or more policies. The one or more policies specify, for example, one or more data transmission requirements comprising, for example, latency, bandwidth, security and/or data gravity. The one or more network slices comprise, for example, a URLLC network slice and/or an eMBB network slice.

(64) In one or more embodiments, at least some of the respective ones of the plurality of data portions from the respective data stores are aggregated based at least in part on one or more common parameters of the one or more parameters. Access by at least one user to the aggregated

ones of the plurality of data portions from the respective data stores is enabled. The aggregating may be performed in response to one or more queries for the data. In some embodiments, a visualization of the aggregated ones of the plurality of data portions is generated, wherein the aggregated ones of the plurality of data portions are received from the respective data stores in a plurality of data streams.

(65) In accordance with illustrative embodiments, access to at least one user to at least some of the respective ones of the plurality of data portions from the respective data stores is assigned based at least in part on a role of the at least one user. At least some of the respective ones of the plurality of data portions from the respective data stores are aggregated based at least in part on the assigned access.

(66) The process may also comprise classifying the data into one or more functional groups, and segregating the data into the plurality of data portions based at least in part on the one or more functional groups.

(67) It is to be appreciated that the FIG. 6 process and other features and functionality described above can be adapted for use with other types of information systems configured to execute data access services in a data management platform or other type of platform.

(68) The particular processing operations and other system functionality described in conjunction with the flow diagram of FIG. 6 are therefore presented by way of illustrative example only, and should not be construed as limiting the scope of the disclosure in any way. Alternative embodiments can use other types of processing operations. For example, the ordering of the process steps may be varied in other embodiments, or certain steps may be performed at least in part concurrently with one another rather than serially. Also, one or more of the process steps may be repeated periodically, or multiple instances of the process can be performed in parallel with one another.

(69) Functionality such as that described in conjunction with the flow diagram of FIG. 6 can be implemented at least in part in the form of one or more software programs stored in memory and executed by a processor of a processing device such as a computer or server. As will be described below, a memory or other storage device having executable program code of one or more software programs embodied therein is an example of what is more generally referred to herein as a “processor-readable storage medium.”

(70) Illustrative embodiments of systems with a data management platform as disclosed herein can provide a number of significant advantages relative to conventional arrangements. For example, the data management platform effectively provides a framework for dividing data into different sets based on one or more parameters, such as policies and geographic locations of devices. This segregation of data advantageously allows for granular control of operations by various entities on data sets and their contents. The management and organization of the data according to the illustrative embodiments facilitates access to the data so that the data can be processed at the edge of a network. The processing of the data at edge locations results in reduced data movement and cost, higher data security and increased data sovereignty. The embodiments further facilitate compliance with local laws for data gravity and reporting.

(71) As an additional advantage, the embodiments further provide for secure access to various data sets based on defined policies. The classification of various data sets into functional groups further facilitates the division of data and how and by whom it is accessed. For example, various users are classified into the functional groups, with each functional group having certain rights to access, modify and/or retrieve the data. Thus, the operational data sets can be secured based on policies defined for each user.

(72) The embodiments also provide technical solutions to map data from edge locations to core/backend computing sites over network slices based on the policies identified during data segregation. Such policies correspond to how the data is to be transmitted so as to be compliant with enterprise and user requirements and agreements governing factors like security, bandwidth

and latency. The embodiments are scalable to keep pace with the increasing number of devices and processing at edge locations, and provide real-time data and corresponding insights to users at core/backend computing sites that need to access this data from distributed data stores at edge locations.

(73) It is to be appreciated that the particular advantages described above and elsewhere herein are associated with particular illustrative embodiments and need not be present in other embodiments. Also, the particular types of information processing system features and functionality as illustrated in the drawings and described above are exemplary only, and numerous other arrangements may be used in other embodiments.

(74) As noted above, at least portions of the information processing system **100** may be implemented using one or more processing platforms. A given such processing platform comprises at least one processing device comprising a processor coupled to a memory. The processor and memory in some embodiments comprise respective processor and memory elements of a virtual machine or container provided using one or more underlying physical machines. The term “processing device” as used herein is intended to be broadly construed so as to encompass a wide variety of different arrangements of physical processors, memories and other device components as well as virtual instances of such components. For example, a “processing device” in some embodiments can comprise or be executed across one or more virtual processors. Processing devices can therefore be physical or virtual and can be executed across one or more physical or virtual processors. It should also be noted that a given virtual device can be mapped to a portion of a physical one.

(75) Some illustrative embodiments of a processing platform that may be used to implement at least a portion of an information processing system comprise cloud infrastructure including virtual machines and/or container sets implemented using a virtualization infrastructure that runs on a physical infrastructure. The cloud infrastructure further comprises sets of applications running on respective ones of the virtual machines and/or container sets.

(76) These and other types of cloud infrastructure can be used to provide what is also referred to herein as a multi-tenant environment. One or more system elements such as the data management platform **110** or portions thereof are illustratively implemented for use by tenants of such a multi-tenant environment.

(77) As mentioned previously, cloud infrastructure as disclosed herein can include cloud-based systems. Virtual machines provided in such systems can be used to implement at least portions of one or more of a computer system and a data management platform in illustrative embodiments. These and other cloud-based systems in illustrative embodiments can include object stores.

(78) Illustrative embodiments of processing platforms will now be described in greater detail with reference to FIGS. **7** and **8**. Although described in the context of system **100**, these platforms may also be used to implement at least portions of other information processing systems in other embodiments.

(79) FIG. **7** shows an example processing platform comprising cloud infrastructure **700**. The cloud infrastructure **700** comprises a combination of physical and virtual processing resources that may be utilized to implement at least a portion of the information processing system **100**. The cloud infrastructure **700** comprises multiple virtual machines (VMs) and/or container sets **702-1**, **702-2**, . . . **702-L** implemented using virtualization infrastructure **704**. The virtualization infrastructure **704** runs on physical infrastructure **705**, and illustratively comprises one or more hypervisors and/or operating system level virtualization infrastructure. The operating system level virtualization infrastructure illustratively comprises kernel control groups of a Linux operating system or other type of operating system.

(80) The cloud infrastructure **700** further comprises sets of applications **710-1**, **710-2**, . . . **710-L** running on respective ones of the VMs/container sets **702-1**, **702-2**, . . . **702-L** under the control of the virtualization infrastructure **704**. The VMs/container sets **702** may comprise respective VMs,

respective sets of one or more containers, or respective sets of one or more containers running in VMs.

(81) In some implementations of the FIG. 7 embodiment, the VMs/container sets **702** comprise respective VMs implemented using virtualization infrastructure **704** that comprises at least one hypervisor. A hypervisor platform may be used to implement a hypervisor within the virtualization infrastructure **704**, where the hypervisor platform has an associated virtual infrastructure management system. The underlying physical machines may comprise one or more distributed processing platforms that include one or more storage systems.

(82) In other implementations of the FIG. 7 embodiment, the VMs/container sets **702** comprise respective containers implemented using virtualization infrastructure **704** that provides operating system level virtualization functionality, such as support for Docker containers running on bare metal hosts, or Docker containers running on VMs. The containers are illustratively implemented using respective kernel control groups of the operating system.

(83) As is apparent from the above, one or more of the processing modules or other components of system **100** may each run on a computer, server, storage device or other processing platform element. A given such element may be viewed as an example of what is more generally referred to herein as a “processing device.” The cloud infrastructure **700** shown in FIG. 7 may represent at least a portion of one processing platform. Another example of such a processing platform is processing platform **800** shown in FIG. 8.

(84) The processing platform **800** in this embodiment comprises a portion of system **100** and includes a plurality of processing devices, denoted **802-1**, **802-2**, **802-3**, . . . **802-K**, which communicate with one another over a network **804**.

(85) The network **804** may comprise any type of network, including by way of example a global computer network such as the Internet, a WAN, a LAN, a satellite network, a telephone or cable network, a cellular network, a wireless network such as a WiFi or WiMAX network, or various portions or combinations of these and other types of networks.

(86) The processing device **802-1** in the processing platform **800** comprises a processor **810** coupled to a memory **812**. The processor **810** may comprise a microprocessor, a microcontroller, an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), a central processing unit (CPU), a graphical processing unit (GPU), a tensor processing unit (TPU), a video processing unit (VPU) or other type of processing circuitry, as well as portions or combinations of such circuitry elements.

(87) The memory **812** may comprise random access memory (RAM), read-only memory (ROM), flash memory or other types of memory, in any combination. The memory **812** and other memories disclosed herein should be viewed as illustrative examples of what are more generally referred to as “processor-readable storage media” storing executable program code of one or more software programs.

(88) Articles of manufacture comprising such processor-readable storage media are considered illustrative embodiments. A given such article of manufacture may comprise, for example, a storage array, a storage disk or an integrated circuit containing RAM, ROM, flash memory or other electronic memory, or any of a wide variety of other types of computer program products. The term “article of manufacture” as used herein should be understood to exclude transitory, propagating signals. Numerous other types of computer program products comprising processor-readable storage media can be used.

(89) Also included in the processing device **802-1** is network interface circuitry **814**, which is used to interface the processing device with the network **804** and other system components, and may comprise conventional transceivers.

(90) The other processing devices **802** of the processing platform **800** are assumed to be configured in a manner similar to that shown for processing device **802-1** in the figure.

(91) Again, the particular processing platform **800** shown in the figure is presented by way of

example only, and system **100** may include additional or alternative processing platforms, as well as numerous distinct processing platforms in any combination, with each such platform comprising one or more computers, servers, storage devices or other processing devices.

(92) For example, other processing platforms used to implement illustrative embodiments can comprise converged infrastructure.

(93) It should therefore be understood that in other embodiments different arrangements of additional or alternative elements may be used. At least a subset of these elements may be collectively implemented on a common processing platform, or each such element may be implemented on a separate processing platform.

(94) As indicated previously, components of an information processing system as disclosed herein can be implemented at least in part in the form of one or more software programs stored in memory and executed by a processor of a processing device. For example, at least portions of the functionality of one or more elements of the data management platform **110** as disclosed herein are illustratively implemented in the form of software running on one or more processing devices.

(95) It should again be emphasized that the above-described embodiments are presented for purposes of illustration only. Many variations and other alternative embodiments may be used. For example, the disclosed techniques are applicable to a wide variety of other types of information processing systems and data management platforms. Also, the particular configurations of system and device elements and associated processing operations illustratively shown in the drawings can be varied in other embodiments. Moreover, the various assumptions made above in the course of describing the illustrative embodiments should also be viewed as exemplary rather than as requirements or limitations of the disclosure. Numerous other alternative embodiments within the scope of the appended claims will be readily apparent to those skilled in the art.

## Claims

1. A method comprising: receiving data from a plurality of devices; analyzing the data to identify one or more parameters for segregation of the data; tagging the data with one or more metadata tags corresponding to the one or more parameters; segregating the data into a plurality of data portions based at least in part on the one or more parameters; and storing respective ones of the plurality of data portions in respective data stores of a plurality of data stores; wherein the one or more parameters comprise at least one of one or more policies corresponding to the data and respective locations of the plurality of devices; wherein the steps of the method are executed by a processing device operatively coupled to a memory.
2. The method of claim 1 further comprising mapping the plurality of data portions to one or more network slices of at least one of a fifth generation (5G) network and a private long-term evolution (pLTE) network based at least in part on the one or more policies.
3. The method of claim 2 wherein the one or more policies specify one or more data transmission requirements.
4. The method of claim 3 wherein the one or more data transmission requirements comprise at least one of latency, bandwidth, security and data gravity.
5. The method of claim 2 wherein the one or more network slices comprise at least one of an ultra-reliable low latency connection (URLLC) network slice and an enhanced mobile broadband (eMBB) network slice.
6. The method of claim 1 wherein at least a subset of the respective data stores are in respective edge locations corresponding to the plurality of devices.
7. The method of claim 1 wherein the data comprises operational data of the plurality of devices.
8. The method of claim 1 further comprising aggregating at least some of the respective ones of the plurality of data portions from the respective data stores based at least in part on one or more common parameters of the one or more parameters.



9. The method of claim 8 further comprising enabling access by at least one user to the aggregated ones of the plurality of data portions from the respective data stores.
10. The method of claim 8 wherein the aggregating is performed in response to one or more queries for the data.
11. The method of claim 8 further comprising generating a visualization of the aggregated ones of the plurality of data portions, wherein the aggregated ones of the plurality of data portions are received from the respective data stores in a plurality of data streams.
12. The method of claim 1 further comprising: assigning access to at least one user to at least some of the respective ones of the plurality of data portions from the respective data stores based at least in part on a role of the at least one user; and aggregating the at least some of the respective ones of the plurality of data portions from the respective data stores based at least in part on the assigned access.
13. The method of claim 1 further comprising: classifying the data into one or more functional groups; and segregating the data into the plurality of data portions based at least in part on the one or more functional groups.
14. An apparatus comprising: a processing device operatively coupled to a memory and configured: to receive data from a plurality of devices; to analyze the data to identify one or more parameters for segregation of the data; to tag the data with one or more metadata tags corresponding to the one or more parameters; to segregate the data into a plurality of data portions based at least in part on the one or more parameters; and to store respective ones of the plurality of data portions in respective data stores of a plurality of data stores; wherein the one or more parameters comprise at least one of one or more policies corresponding to the data and respective locations of the plurality of devices.
15. The apparatus of claim 14 wherein the processing device is further configured to map the plurality of data portions to one or more network slices of at least one of a fifth generation (5G) network and a private long-term evolution (pLTE) network based at least in part on the one or more policies.
16. The apparatus of claim 14 wherein the processing device is further configured to aggregate at least some of the respective ones of the plurality of data portions from the respective data stores based at least in part on one or more common parameters of the one or more parameters.
17. The apparatus of claim 14 wherein the processing device is further configured: to assign access to at least one user to at least some of the respective ones of the plurality of data portions from the respective data stores based at least in part on a role of the at least one user; and to aggregate the at least some of the respective ones of the plurality of data portions from the respective data stores based at least in part on the assigned access.
18. An article of manufacture comprising a non-transitory processor-readable storage medium having stored therein program code of one or more software programs, wherein the program code when executed by at least one processing device causes said at least one processing device to perform the steps of: receiving data from a plurality of devices; analyzing the data to identify one or more parameters for segregation of the data; tagging the data with one or more metadata tags corresponding to the one or more parameters; segregating the data into a plurality of data portions based at least in part on the one or more parameters; and storing respective ones of the plurality of data portions in respective data stores of a plurality of data stores; wherein the one or more parameters comprise at least one of one or more policies corresponding to the data and respective locations of the plurality of devices.
19. The article of manufacture of claim 18 wherein the program code further causes said at least one processing device to map the plurality of data portions to one or more network slices of at least one of a fifth generation (5G) network and a private long-term evolution (pLTE) network based at least in part on the one or more policies.
20. The article of manufacture of claim 18 wherein the program code further causes said at least

one processing device to aggregate at least some of the respective ones of the plurality of data portions from the respective data stores based at least in part on one or more common parameters of the one or more parameters.

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